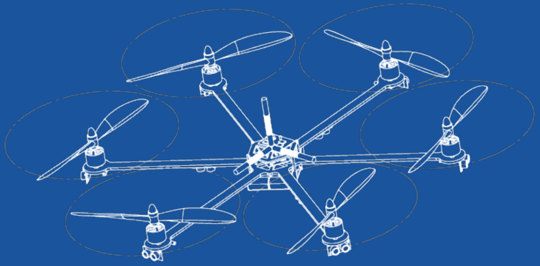
Universität Stuttgart
Institut für Flugmechanik und Flugregelung

Region of attraction with integral quadratic constraints

System Theoretical Methods for Flight Control

Saturday 16th May, 2026

First Last



Outline

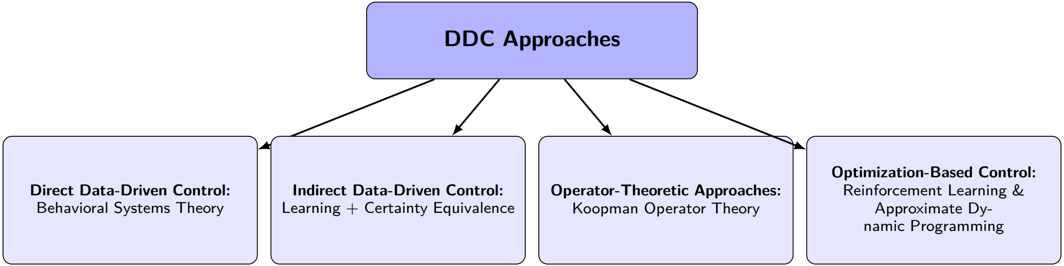
Today's Agenda

1. Introduction

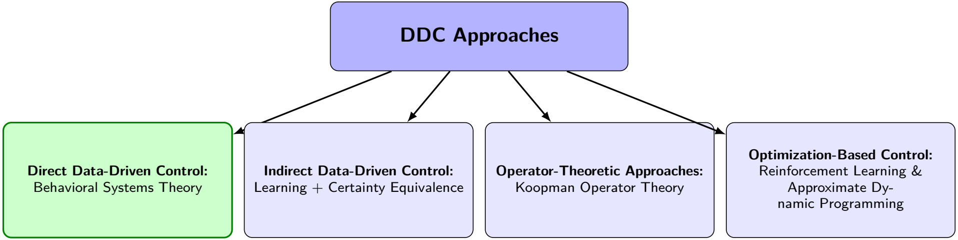
Motivation for Data-Driven Control

- **Bypassing System Identification:** Directly design controllers from experimental data without needing a highly accurate parametric model.
- **Handling Complexity:** Highly effective for systems with dynamics that are notoriously difficult to model from first principles (e.g., complex aerodynamics, friction).
- **Reducing Modeling Errors:** Mitigates performance and safety degradation caused by unmodeled dynamics or model-reality mismatches.
- **Time and Cost Efficiency:** Saves the significant engineering time and effort usually required to build and validate mathematical models.

Different Approaches for Data-Driven Control



Different Approaches for Data-Driven Control



- **Specific Methods Used in This Paper for Robustness:**
  - **Density Functions:** Serve as safety certificates to prove the system avoids unsafe regions.
  - **Bounded Process Noise:** Explicitly accounts for unknown but bounded experimental noise/disturbances.
  - **Convex Optimization (SDP):** Reformulates non-convex safety constraints into tractable Semidefinite Programs.
  - **Sum-of-Squares (SOS):** Relaxes polynomial inequalities to solve the SDP computationally.

Goal of the Paper

Nonlinear System Setup:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t)) + g(\mathbf{x}(t))\mathbf{u}(t) + \mathbf{w}(t)$$

- State $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^n$, Input $\mathbf{u} \in \mathcal{U} \subset \mathbb{R}^m$
- Unknown true dynamics $f(\mathbf{x})$ and $g(\mathbf{x})$ (only noisy data is available)
- Unknown but bounded process noise/disturbance $\mathbf{w}(t) \in \mathcal{W}$

Intended Results (Robust Safety Objective)

Synthesize a state feedback control law $\mathbf{u} = k(\mathbf{x})$ directly from data such that:

- The closed-loop system is **robustly safe**.
- Trajectories starting in an initial set $\mathcal{X}_0$ **never** enter an unsafe region $\mathcal{X}_u$, despite any allowable disturbance $\mathbf{w} \in \mathcal{W}$.

Today's Agenda

1. Introduction

2. Traditional Approach & its Challenges

Traditional Approach

Traditionally, robustness and safety verification are achieved through barrier certificates [1].

Barrier Functions for Safety:

- A barrier function $B(\mathbf{x})$ separates safe states from unsafe states $\mathcal{X}_u$.
- **Condition:** $B(\mathbf{x}) > 0$ for $\mathbf{x} \in \mathcal{X}_u$ and $B(\mathbf{x}) \leq 0$ for safe initial states $\mathcal{X}_0$.
- To ensure trajectories never enter $\mathcal{X}_u$, the derivative along the system dynamics must satisfy:

$$\dot{B}(\mathbf{x}) = \frac{\partial B(\mathbf{x})}{\partial \mathbf{x}} \left(f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} + \mathbf{w} \right) \leq 0 \quad \text{on the boundary of the safe set}$$

The Challenge

Jointly searching for a barrier certificate $B(\mathbf{x})$ and a control policy $\mathbf{u} = k(\mathbf{x})$ from data inherently results in a **non-convex optimization problem** (due to the multiplication of the two unknowns).

Today's Agenda

1. Introduction
2. Traditional Approach & its Challenges
3. Proposed Solution & its Advantage

Proposed Solution

To overcome the non-convexity of barrier functions, the proposed framework uses **Density Functions** $\rho(\mathbf{x})$ as an alternative safety certificate, which was proposed in [2].

Key Idea (Dual to Lyapunov/Barrier):

- Instead of preventing trajectories from crossing a strict boundary, density functions prove that the "flow" of trajectories diverges away from the unsafe set $\mathcal{X}_u$.
- Safety is guaranteed by verifying a linear partial differential inequality on $\rho(\mathbf{x})$.

Convexifying the Synthesis Problem

By introducing a change of variables $\mathbf{v}(\mathbf{x}) = \rho(\mathbf{x})\mathbf{u}(\mathbf{x})$, the bilinear coupling of the unknown certificate and the unknown controller becomes **linear**.

$$\rho(\mathbf{x})\mathbf{u}(\mathbf{x}) \text{ (non-convex)} \quad \longrightarrow \quad \mathbf{v}(\mathbf{x}) \text{ (convex)}$$

Deriving the SDP Formulation

Since the true continuous-time dynamics are unknown, we rely directly on the collected noisy experimental data and the known noise bounds.

From Data to Semidefinite Programming (SDP):

- **Polynomial Assumptions:** The system dynamics, density function $\rho(\mathbf{x})$, and controller $\mathbf{v}(\mathbf{x})$ are parameterized as polynomials.
- **Robust Constraints:** The safety condition must hold for all possible disturbances $\mathbf{w} \in \mathcal{W}$ that are consistent with the data.
- **Sum-of-Squares (SOS) Relaxation:** The strict polynomial non-negativity conditions are relaxed into SOS conditions, which are computationally tractable.

Resulting Optimization

The robust safety constraints are reformulated into Linear Matrix Inequalities (LMIs) using the data. The entire synthesis problem is solved as a **Semidefinite Program (SDP)**.

Advantages Over Traditional Approaches

The proposed density-function framework provides several distinct advantages:

- **Convex Optimization:** Transforms a notoriously difficult non-convex problem (jointly finding a certificate + controller) into a convex SDP, guaranteeing a global optimum.
- **Tractable Robustness:** Incorporating bounded process noise mathematically translates into robust LMI conditions, cleanly handled by SOS without making the problem unsolvable.
- **Directly Data-Driven:** Sidesteps the need to identify the exact nonlinear model first, which is often highly sensitive to modeling errors.
- **Rational Controller:** The resulting state feedback law $\mathbf{u}(\mathbf{x}) = \frac{\mathbf{v}(\mathbf{x})}{\rho(\mathbf{x})}$ is a rational function, allowing for highly flexible and expressive control policies.

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1. Introduction
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3. Proposed Solution & its Advantage
4. Validation

Validation

Experimental verification and performance analysis

Today's Agenda

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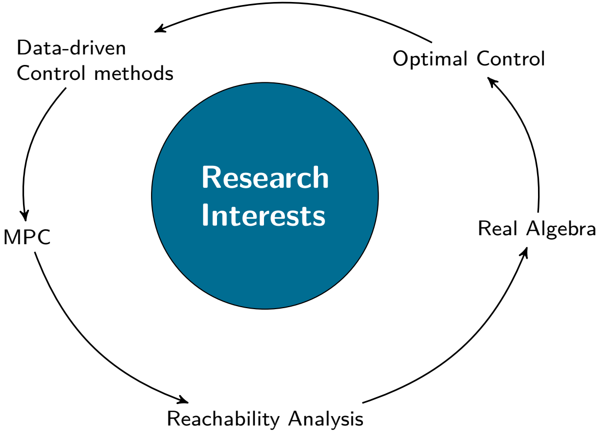
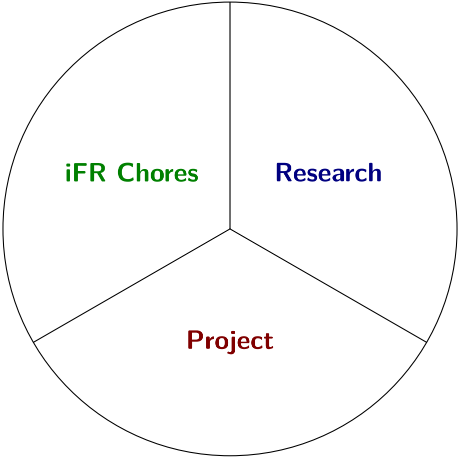
3. Proposed Solution & its Advantage
4. Validation
5. TikZ Examples

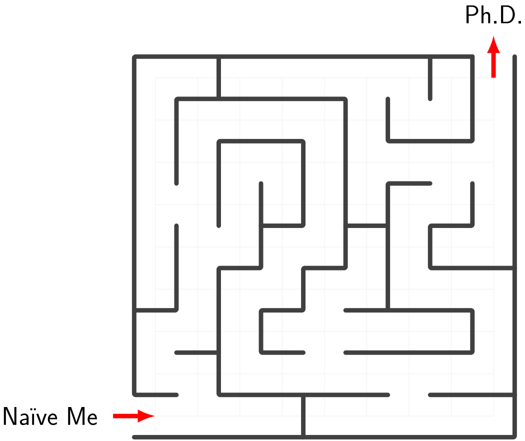
In the next slides, we will see some examples.


Data-driven
Control methods
Optimal Control
Research
Interests
MPC
Real Algebra
Reachability Analysis

TikZ Example - 2



TikZ Example - 3



TikZ Example - 4

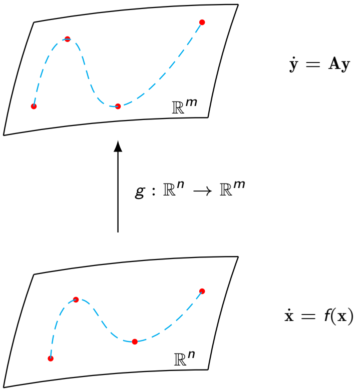


Figure 1: Some description

TikZ Keviats

Well, tikz is AWESOME, however as you probably noticed by now, it can take a lot of time compiling the whole document if it contains multiple tikzpictures.
If possible, generate the tikzpictures in a standalone file, i.e., generate .pdf files with the figures and only import those to the main file. Believe me, it will reduce by a lot the time you spend waiting for the compiled file.

Today's Agenda

Equations

Since this is latex, we can use equations.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{1}$$

All equations are numbered by the order they appear in the presentation. But if you don't want a numbered equation, you can still do it

$$E = mc^2$$

Blocks

One useful and nice feature is the block environment (*newtheorem*). It can be used to highlight important information. (see preamble.tex)

Conferences

1. *American Control Conference, Toronto, July 2024*
   Initial deadline: ***September 22, 2023***
2. *European Control Conference, Stockholm, June 2024*
   Initial deadline: ***October 25, 2023***

Theorem (Theorem name)

This is a theorem.

Remark

This is a remark.

Include figures

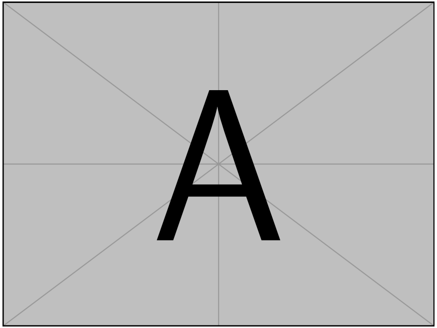


Figure 2: Caption 1

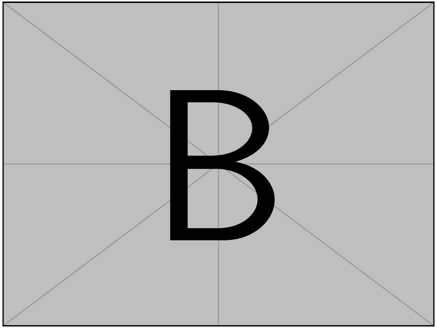
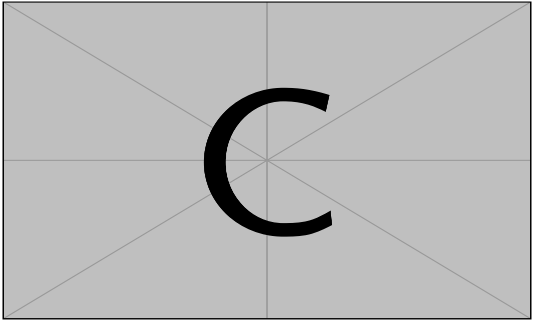


Figure 3: Caption 2

Insert videos



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Citations

Just as a normal .tex document we can use .bib files to cite references.

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The references are shown at the end of the presentation automatically.

Bibliography

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In the current state you just need to change the photo and a few information details.

Universität Stuttgart
Institute of Flight Mechanics and Controls

Thank you!

First Last

E-Mail: ...@ifr.uni-stuttgart.de

References I

[1] S. Prajna and A. Jadbabaie. Safety verification of hybrid systems using barrier certificates. In *Proc. HSCC*, volume 2993, pages 477–492, 2004.

[2] A. Rantzer and S. Prajna. On analysis and synthesis of safe control laws. In *Proc. 42nd Allerton Conf. Commun. Control Comput.*, pages 1468–1476, 2004.